

QU!D PROTOCOL

Cross-Chain Prediction Market Infrastructure
with Confidence-Weighted Settlement

"We detect stress. We don't predict it."

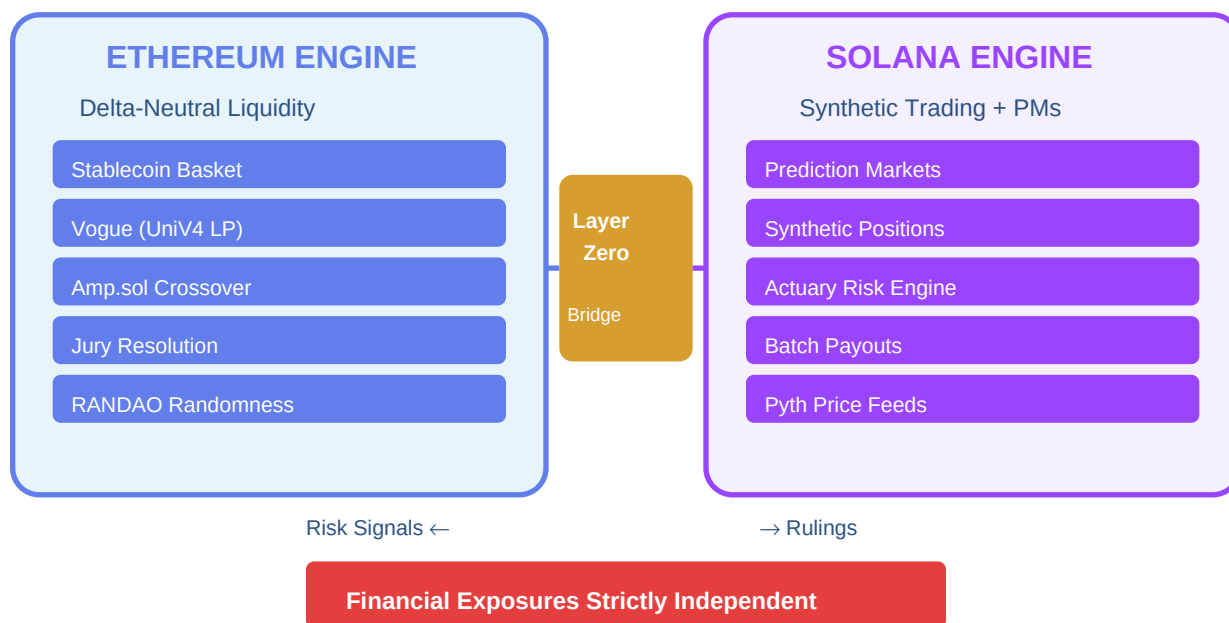


Figure 1: Dual-Engine Architecture

Abstract

QUID Protocol introduces a novel cross-chain prediction market infrastructure that bridges Solana's throughput capabilities with Ethereum's deep liquidity and composability. The protocol features confidence-weighted settlement, a dual-payout mechanism that separates trading profits from forecasting accuracy rewards, and a jury-based resolution system secured by staked liquidity.

By leveraging prediction market signals for stablecoin risk assessment, QUID creates a self-reinforcing ecosystem where market intelligence directly improves collateral management. The symbiotic relationship between Solana's synthetic trading demand and Ethereum's liquidity provision creates powerful network effects that benefit all participants.

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Dual-Engine Architecture

QU!D is not a single protocol—it is a **dual-engine machine**. The two engines share a brand and a cross-chain 'Truth' bridge, but their financial exposures are strictly independent.

Engine	Chain	Function	Financial Primitive
Liquidity Engine	Ethereum	Delta-neutral LP provision	Stablecoin basket, yield strategies
Trading Engine	Solana	Leverage + prediction markets	Synthetic positions, forecasting

Independence Principle

A critical design choice: **Ethereum positions never settle against Solana losses, and vice versa**. Solana generates 'Truth' about stablecoin depegs. This signal travels via LayerZero to Ethereum for fee adjustment. But treasuries are isolated—a black swan on one chain cannot drain the other.

The Symbiotic Relationship

The engines aren't just parallel—they're symbiotic. Demand for Solana synthetic trading forces users to deposit stables on Ethereum first (then bridge). This creates natural TVL growth without artificial incentives:

Solana trading demand → Users need collateral → Must deposit on Ethereum → Basket TVL grows → More yield for LPs → More deposits → Deeper Solana collateral → More trading capacity

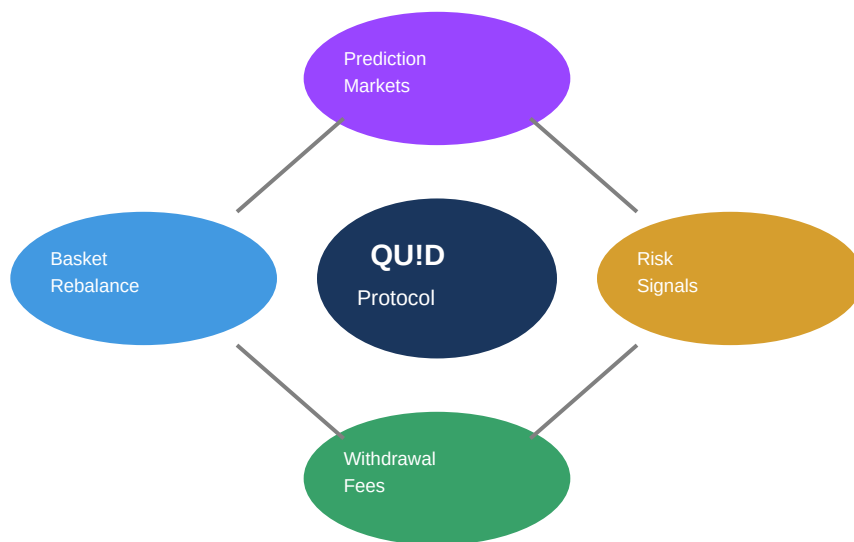


Figure 2: Closed-Loop Signal Flow

Cross-Chain Design

The cross-chain design is not complexity for its own sake—it's a fundamental requirement driven by irreconcilable constraints between throughput and liquidity.

Why Solana for Trading

- Sub-second block times enable responsive trading
- Complex payout calculations (confidence weighting, time decay) execute cheaply
- Batch reveal and payout distribution for many positions atomically
- High-frequency position updates at scale

Why Ethereum for Settlement

- Deep stablecoin liquidity across multiple tokens
- Composable DeFi infrastructure: Uniswap V4, Morpho, AAVE integration
- RANDAO provides unbiased randomness for jury selection
- Institutional preference for ETH-denominated yield

LayerZero V2 Messages

Direction	Message Type	Purpose
SOL → ETH	RESOLUTION_REQUEST	Market deadline reached, initiate jury
SOL → ETH	JURY_COMPENSATION	Fees + DepegStats for risk signals
ETH → SOL	FINAL_RULING	Winning sides, slashing addresses

Cross-Chain Learning

The dual-chain architecture enabled cross-pollination of design patterns:

- **Solana → Ethereum:** The 'discriminator' pattern from Solana was adapted for Vogue's internal accounting
- **Ethereum → Solana:** UniswapV4's credit/debit sign convention informed stay.rs and clutch.rs

Risk-Responsive Fee Mechanism

The core insight: **withdrawal fees should reflect forward-looking risk, not backward-looking price data.** Traditional baskets use fixed fees that subsidize risk-takers. When one constituent shows stress, governance votes are too slow and oracle triggers are reactive.

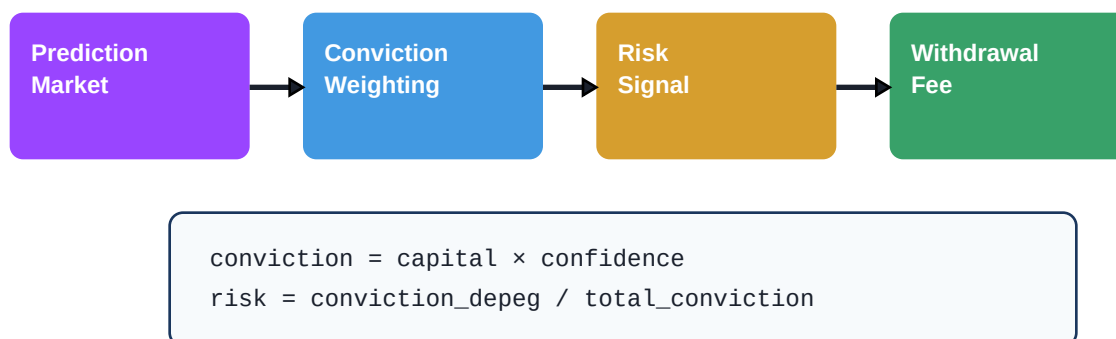


Figure 3: Risk Signal Flow

The Inverse Relationship

- **Higher perceived risk** → **Lower withdrawal fee** (incentivize removal of risky assets)
- **Lower perceived risk** → **Higher withdrawal fee** (discourage cherry-picking safe assets)

This creates a self-correcting system. When sophisticated traders detect early depeg signals, they face reduced friction to exit that asset. Their exit improves basket composition without governance intervention.

Conviction Weighting

The risk signal comes from prediction markets, but not raw capital flows. QU!D uses conviction weighting:

$$\text{conviction} = \text{capital} \times \text{confidence}$$

A \$10,000 position at 90% confidence carries more weight than \$15,000 at 50% confidence. This filters noise from low-conviction speculation and amplifies signals from informed traders.

The Pieralberto Formula

$$\text{Fee} = \text{Base (4 bps)} + \text{Absolute Premium (risk} \times 0.4) + \text{Relative Spread (position}^2 \times \text{intensity)}$$

Average fee: 5.9 bps (1.47× Curve's baseline). The quadratic position² scaling creates strong incentives at the margin.

Design Philosophy: The fee mechanism *detects* stress—it doesn't *predict* losses. We price protocol costs of facilitating trades, not expected losses. Monotonic pricing (riskier = costlier) steers behavior correctly without requiring precise probability estimates. Observable proxies beat distributional assumptions.

Prediction Market Design

Adaptive LMSR

Unlike constant-product AMMs ($x \cdot y = k$), QUID uses a modified Logarithmic Market Scoring Rule (LMSR) adapted for dynamic liquidity. The parameter b is not static—it adjusts based on market conditions:

$$b^* = b_{\text{min}} \times \text{time_factor} \times \text{volume_factor} \times \text{complexity_factor}$$

- **time_factor** = $\ln(1 + \text{age_in_days})$ — markets become more liquid as they mature
- **volume_factor** = $(\text{actual_volume} / \text{expected_volume})^{0.25}$ — active markets get deeper liquidity
- **complexity_factor** = $n^{0.25} \times (1 + \ln(n)/10)$ — more outcomes need deeper books

Information Decay (λ)

The protocol assumes information value decays over time. Adaptive λ controls the decay curve:

- **Short duration / High complexity:** λ is high $\rightarrow$ early traders rewarded significantly more
- **Long duration:** λ decreases $\rightarrow$ system values late-stage consensus aggregation

λ ranges from 0.5 to 10.0. This rewards commitment: traders who identify correct outcomes early and hold through volatility receive proportionally higher payouts.

TWAP Manipulation Resistance

Orders are validated against Time-Weighted Average Price. If spot price deviates more than threshold (default 3%) from TWAP, the order is rejected. This doesn't prevent all manipulation, but forces attackers to sustain manipulated prices over time—dramatically increasing capital requirements and risk.

Keeper-Delegated Auto-Reveal

Users can designate a `reveal_delegate`—a keeper address authorized to reveal on their behalf. The keeper monitors markets, reveals for all delegating users when the window opens, and earns a small fee. This eliminates UX friction: set-and-forget. Self-reveal always available for those who prefer it.

Two-Payout Mechanism

Traditional prediction markets force a binary choice: exit or stay. QU!D introduces a fundamental innovation—**two distinct paths to profit**.

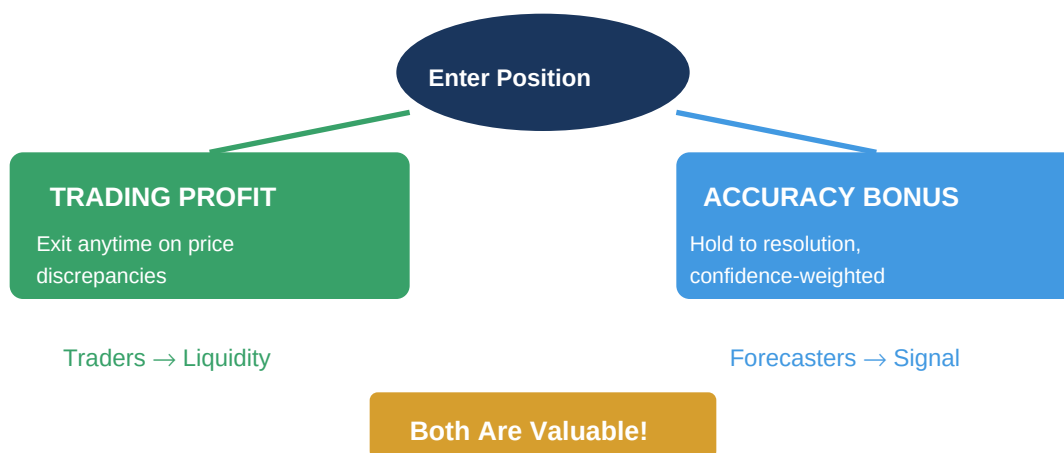


Figure 4: Market Segmentation

Path 1: Trading Profit (Exit Early)

Traders profit from price discrepancies during the market's lifetime. A trader who buys at \$0.30 and sells at \$0.50 captures \$0.20 per share—regardless of final outcome. They're rewarded for identifying mispricings.

$$\text{Trading Profit} = (\text{Exit Price} - \text{Entry Price}) \times \text{Shares}$$

Path 2: Accuracy Bonus (Hold to Resolution)

Forecasters hold through resolution and receive a share of losing capital:

$$\text{Accuracy Payout} = (\text{Loser Capital} \times \text{Your Weight}) / \text{Total Winner Weight}$$

Payout pools: Winner pot = 80% of loser capital; Loser pot = 20% (losers get some back based on weight). This rewards epistemic humility—uncertain losers preserve capital.

Confidence Percentile System

Outcome	Confidence	Percentile	Result
Winner	90%	90th	Maximum share of winnings
Winner	50%	50th	Moderate share
Loser	90%	10th	Severely punished
Loser	50%	50th	Preserves more capital

Table 1: Confidence-Weighted Payout — Knowing What You Don't Know

The Actuary: Dynamic Risk Engine

The Actuary is QU!D's dynamic risk engine that governs synthetic positions. It penalizes or rewards users based on their impact on pool stability.

Asymmetric Volatility Detection

```
observed_vol_bps = max(vol_floor, min(3000, raw_vol))
```

- Volatility can spike to 3000 bps **instantly** (no smoothing on the way up)
- Volatility can **never** go below vol_floor (permanent lower bound)

This creates one-way ratcheting. When stress hits, the system responds immediately. When calm returns, the floor prevents premature relaxation.

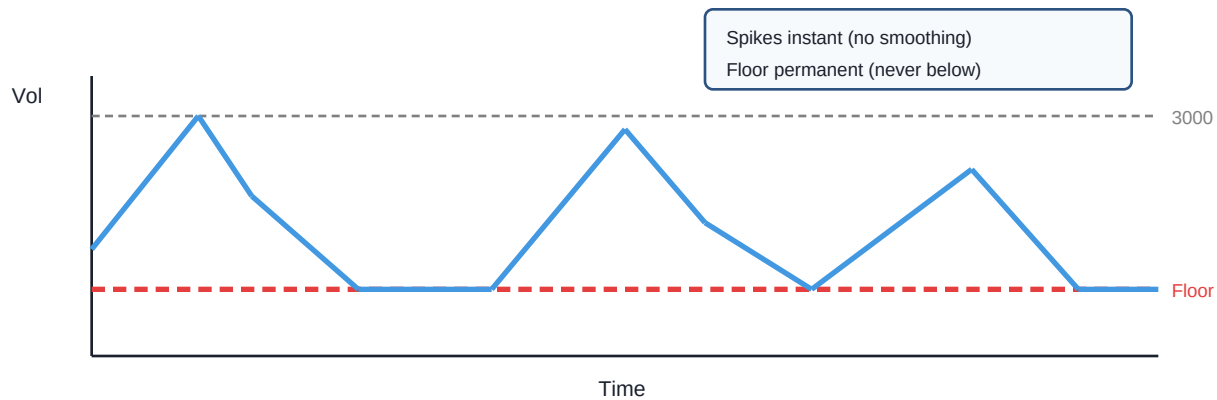


Figure 5: Asymmetric Volatility Protection

Jump Detection: Persistent Memory

Volatility spikes increment `jump_count`. Higher jump counts → higher fees and tighter collars. Critically, **jump_count persists even if the volatility EMA decays**. Historical stress leaves lasting imprint.

Why Symmetric EMA?

Symmetric EMA is easier to audit, predict, and reason about. The asymmetric protection comes from the **floor** and **jump count**, not the EMA itself. Separation of concerns: EMA tracks smooth; floor bounds; jump count remembers.

Directional Risk Detection

Trade Impact	Fee Multiplier	Rationale
Strong hedge	0.7×	Actively improves pool balance
Mild hedge	0.85×	Slightly helps
Neutral	1.0×	No impact on pool direction
Mild concentrate	1.5×	Slightly worsens imbalance
Strong concentrate	3.0×	Significantly increases directional risk

Table 2: Fee Scaling by Pool Impact

Gradual Amortization

Traditional DeFi liquidations are MEV goldmines. When a position becomes undercollateralized: liquidation bots race, block builders reorder, sandwich attacks front-run, and cascades crash prices. Only bots win.

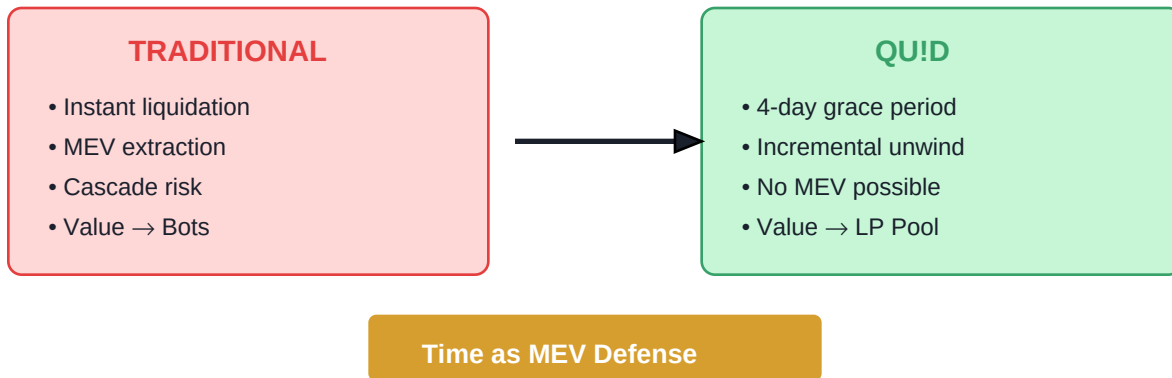


Figure 6: Time as MEV Defense

The Mechanism

When a position breaches its collar:

1. **Grace period:** Position holder has ~4 days to add collateral
2. **Amortization begins:** After grace, exposure reduces incrementally
3. **Continuous unwinding:** Each call reduces position by a fraction
4. **Completion:** Position closes or returns to healthy state

$$\text{speed} = 0.5 + 1.5 \times \text{pool_utilization}$$

$$\text{reduction} = (\text{exposure} \times \text{elapsed} / \text{grace_period}) \times \text{speed}$$

The Collar as Hedge Wrapper

Unlike Ostium (where uncapped gains for borrowers lead to uncapped losses for depositors), QUID's collar caps both gains and losses. A position cannot exceed $\pm X\%$ of pledged collateral. For traders: limited upside, but also limited downside and no sudden liquidation. For depositors: bounded liability per position.

Value Flow

Traditional: Position holder → Liquidation bot → Block builder

QUID: Position holder → LP pool (if loss) or keeps profit (if gain)

The LP pool captures value that would otherwise leak to MEV. Liquidators receive only 0.4% tip.

Ethereum Yield Layer

Amp.sol: The Crossover Strategy

Traditional LP strategies are defensive—they provide liquidity and suffer when the market moves too fast. Amp is offensive—it actively exploits momentum.

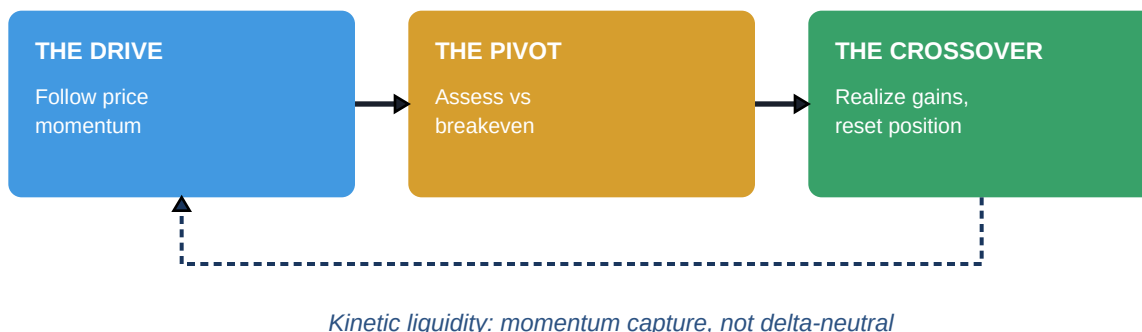


Figure 7: The Crossover Strategy

In basketball, a crossover involves driving in one direction, planting a pivot foot when the defender commits, then shifting weight to the opposite side. Amp performs a financial version using Uniswap V3 price as the 'defender'.

- **The Drive:** Protocol follows trend, throwing collateral in direction of price movement
- **The Pivot:** Assess current value vs breakeven—'plant foot' to evaluate momentum
- **The Crossover:** When momentum sufficient ($\pm 2.5\%$), realize gains, split 50/50 with protocol, reset

Vogue Protocol (UniV4 Liquidity)

- Automated range management—detects drift, atomically repositions
- Batch repositioning—not sandwichable
- Auto-compounding fees on every interaction
- V4 singleton architecture for gas efficiency

Cold Start Solution

Traditional UniV3 forces ETH holders to sell ~50% for USDC—immediate 'delta shedding'. QUID inverts this:

1. **Bonds attract dollars first:** Basket mint provides upfront yield tokens
2. **Virtual USDC:** Vogue creates mockUSD backed by Aux deposits
3. **Full ETH exposure:** ETH depositors maintain upside while earning LP fees

Basket Vault Design

ERC6909 Multi-Token Architecture

QU!D uses ERC6909 to represent various maturities. Each deposit month receives a distinct token ID. Depositors in the same month receive fungible tokens; different months are distinct with independent yields.

Token ID 1 = January 2026 deposits

Token ID 2 = February 2026 deposits

New batches don't dilute old batches. If Batch 1 earns \$3, then Batch 2 joins with \$200, Batch 1 is still redeemable for original + yield. Late arrivals don't capture early yield.

Bond-Like Yield Tokenization

Traditional yield farming requires patience—deposit, wait a year. QU!D introduces **upfront yield tokenization**:

1. Depositor commits capital for a maturity period
2. Protocol calculates expected yield
3. Depositor receives **upfront yield tokens immediately**
4. Tokens tradeable in DeFi, redeemable 1:1 at maturity

Seed investors (13+ month lockup) receive 3× return, funded by 0.25% spread until \$3M collected.

Value to Constituent Stablecoins

QU!D doesn't just aggregate stablecoins—it provides **substantial value back**. Most stablecoins benefit when users stake in native yield vaults (USDS→sUSDS, DAI→sDAI, FRAX→sFRAX). QU!D's maturity schedule (1-12 months) creates additional time-lock incentives on top of native staking. Stablecoin protocols benefit from deeper, stickier liquidity.

Multi-Chain Deployment

Chain	Stablecoins
Ethereum L1	USDC, USDT, DAI, FRAX, USDS, USDE, CRVUSD, PYUSD, GHO
Arbitrum	USDC, USDT, DAI, GHO, FRAX, USDE, USDS, CRVUSD
Base	USDC, USDS, GHO, USDT, DAI, FRAX, USDE, CRVUSD
Polygon	USDC, USDT, DAI, FRAX, CRVUSD
Unichain	USDC, USDT, USDS, FRAX

Jury Resolution System

Market resolution is handled by a jury of staked liquidity providers. Critical distinction: jurors are selected via RANDAO with **uniform random probability**. Stake size does NOT affect selection probability—it only determines slashing exposure.

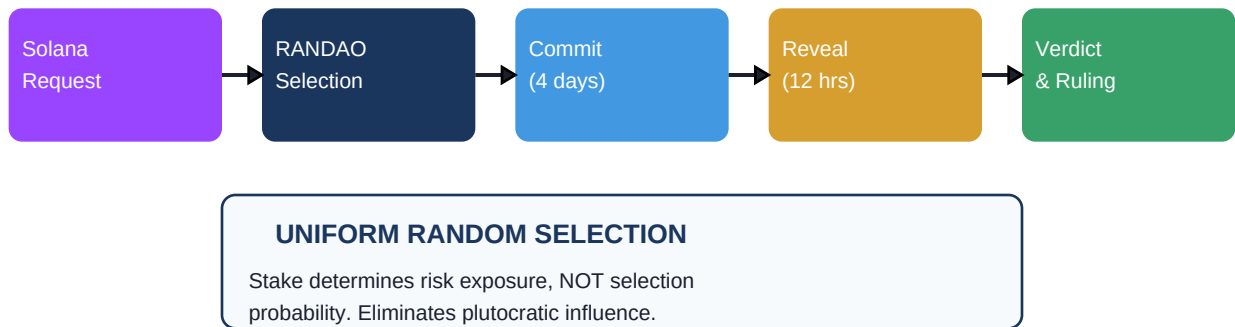


Figure 8: Jury Resolution Flow

Resolution Flow

Step	Location	Action
1. Request	Solana	Market deadline reached, send to Ethereum
2. Voir Dire	Ethereum	RANDAO-based juror selection (21 jurors)
3. Commit	Ethereum	Jurors submit vote hashes (4 days)
4. Reveal	Ethereum	Jurors reveal votes (12 hours)
5. Finalize	Ethereum	Tally votes (2/3 majority), apply slashing
6. Ruling	Solana	Final ruling received, payouts processed

Appeal System

- 7-day appeal window with escalating bonds (1.5× per appeal)
- Maximum 3 appeals, 10 total rounds
- Correct voters split: market fees + slashed stakes from incorrect voters
- Force majeure: 3 hung juries → market cancelled, capital returned

Evidence System (Proof.sol)

Anyone can submit affidavits (Merkle-proven attestations) supporting outcomes. Jurors score evidence as Helpful, Misleading, or Fabricated. Fabricated evidence results in slashing and ban.

Appendix: Technical Summary

Aspect	Traditional DeFi	QU!D Protocol
Liquidations	Instant, MEV-extractable	Gradual (~4 days), MEV-resistant
Prediction Payouts	Capital-only	Confidence-weighted
Risk Signals	Oracle-only	Skin-in-the-game participants
LP Yield	First-come, first-served	Time-weighted (sticky capital)
Basket Rebalancing	Governance votes	Automatic via withdrawal fees
Jury Selection	Token-weighted probability	Uniform random (stake = risk)
Volatility Response	Symmetric	Asymmetric (floor + jump memory)
Cross-Chain	Shared risk	Independent treasuries

Table 3: Key Differentiators

Component	Chain	Key Features
Prediction Markets	Solana	Adaptive LMSR, confidence weighting, TWAP protection
Synthetic Positions	Solana	Leveraged exposure, dynamic collars, gradual amortization
Basket.sol	Ethereum	Multi-stablecoin aggregator, OFT bridge, withdrawal fees
Court.sol	Ethereum	Dispute coordination, appeal handling
Jury.sol	Ethereum	Random selection, staked voting, compensation
Proof.sol	Ethereum	Merkle verification, affidavit tracking
Vogue Protocol	Ethereum	UniV4 LP management, auto-compounding
Amp.sol	Ethereum	Crossover strategy, AAVE integration, momentum capture

Table 4: Component Summary

"The protocol is a facilitator, not a counterparty."

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